321 HW 8
Wednesday, April 8, 2020 4:33 PM

① — Fisher Tropsh —

FT-VIII) Derive rate expression

rev. rxns.

① $CO + * \rightleftharpoons CO^*$ $\quad r_1 = -K_1([CO] - [CO^*])$

② $H_2 + 2* \rightleftharpoons 2H^*$ $\quad r_2 = -K_2([H_2] - [H^*]^2)$

③ $CO^* + 2H^* \rightleftharpoons HCOH^* + 2*$ (RLS) $\quad r_3 = -K_3([CO^*]\cdot[H^*]^2 - [HCOH^*])$

④ $HCOH^* + * \rightleftharpoons CH^* + OH^*$ $\quad r_4 = -K_4([HCOH^*] - ([CH^*]\cdot[OH^*]))$

⑤ $CH^* + H^* \rightleftharpoons CH_2^* + *$ $\quad r_5 = -K_5([CH^*][H^*] - [CH_2^*])$

⑥ $OH^* + H^* \rightleftharpoons H_2O + 2*$ $\quad r_6 = -K_6([OH^*]\cdot[H^*] - [H_2O])$

need $[CO^*]$, $[H^*]$!

Overall: $CO + \cancel{*} + H_2 + 2H^* \rightleftharpoons CH_2^* + H_2O + \cancel{2*}$

$$\boxed{CO + H_2 + 2H^* \rightleftharpoons CH_2^* + * + H_2O}$$

$$\frac{r_1}{K_1} = \frac{r_2}{K_2} = \frac{r_4}{K_4} = \frac{r_5}{K_5} = \frac{r_6}{K_6} = 0 \text{ as Rxn 3 is RLS.}$$

MATERIAL BALANCES.

① $0 = [CO] - [CO^*] \rightarrow [CO] = [CO^*]$

② $[H_2] = [H^*]^2$

④ $[CH^*][OH^*] = [HCOH^*]$

⑤ $[CH^*][H^*] = [CH_2^*]$

⑥ $[OH^*][H^*] = [H_2O]$

③ $-r_3 = K_3[[CO^*][H^*] - [HCOH^*])$

$$\underbrace{K_1 P_{CO} C_V - K_1 (C_{CO^*})}_{} = \underbrace{K_2 (P_{H_2})(C_V)^2 - K_2 (C_{H^*})^2}_{}$$

$$\frac{r_1}{K_1} = 0 = P_{CO} C_V = C_{CO^*} \qquad (P_{H_2} C_V^2 - C_{H^*}^2)\cancel{K_2} = \frac{-r_2}{K_2} = 0$$

$$\boxed{C_{CO^*} = P_{CO} C_V \cdot K_1 = -r_1} \qquad C_{H^*}^2 = P_{H_2} C_V^2 = 0$$

$$\sqrt{C_{H^*}^2} = \sqrt{P_{H_2} C_V^2 K_2} = -r_2$$

$$C_{H^*} = C_V \sqrt{P_{H_2} K_2}$$

$$\boxed{C_{H^*} = C_V K_2} \qquad \text{set } K_2 = \sqrt{P_{H_2} K_2}$$

Site Balance

$$C_T = C_V + C_{CO^*} + C_{H^*}$$

plug in ↓

$$C_T = C_V + P_{CO} C_V \cdot K_1 + C_V K_2$$

$$C_T = C_V(1 + P_{CO} K_1 + K_2)$$

$$C_V = \frac{C_T}{(1 + P_{CO} K_1 + K_2)}$$

?

Plug in Cv

(10-26)

$$\boxed{-r_3 = K_3 (P_{CO})(C_V) K_1 \cdot C_V \cdot K_2 - \frac{C_{HCOH^*} P_*^2}{K_3}}$$

② Rxn occurs on the surface of a catalyst

$$2A_{(g)} + D_{(g)} \rightarrow B_{(g)} + C_{(g)}$$

overall rate of rxn ↓

$$-r_A = \frac{K P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

Suggest a reaction mechanism + show how you would derive above eqn.

Proposed Mechanism ↓

$2A_{(g)} + 2S \rightleftharpoons 2AS_{(g)}$ Adsorption ①

$-r_{AD} = K_{AD}(P_A C_V - \frac{1}{K_A} C_{A\cdot S})$

$2AS_{(g)} + D_{(g)} \longrightarrow BS_{(g)} + C_{(g)} + S_{(s)}$ Surface Rxn ② RATE Limiting STEP

$r_S = K_S(C_{AS}^2 P_D)$

$BS_{(g)} \rightleftharpoons B_{(g)} + S_{(s)}$ Desorption ③

$-r_D = K_D(C_{BS} - K_B P_B C_V)$

Here,

$$\frac{r_{AD}}{K_{AD}} = \frac{r_D}{K_D} = 0 \quad *$$

$-r_D = K_D(C_{BS} - K_B P_B C_V)$

Here,

$$\frac{r_{AD}}{K_{AD}} = \frac{r_D}{K_D} = 0 \quad *$$

① $0 = P_A C_V - \frac{1}{K_A} C_{A\cdot S}$ ADS.

$P_A C_V K_A = C_{A\cdot S}$ ✓

③ $0 = C_{B\cdot S} - K_B P_B C_V$

$P_B C_B C_V = C_{B\cdot S}$ ✓

Site Balance

$C_T = C_V + C_{AS} + C_{BS}$

$C_T = C_V + K_A P_A C_V + K_B P_B C_V$

$C_T = C_V(1 + K_A P_A + K_B P_B)$

$$C_V = \frac{C_T}{(1 + K_A P_A + K_B P_B)}$$

$r_S = K_S (C_{A\cdot S})^2 \cdot P_D$

where $C_{A\cdot S} = K_A P_A C_V$

$r_S = K_S K_A^2 P_A^2 \cdot C_V^2 \cdot P_D$

$$r_S = \frac{K_S K_A^2 P_A^2 C_T^2 P_D}{(1 + K_A P_A + K_B P_B)^2}$$ set $K = K_S K_A^2 C_T^2$

$$\boxed{r_S = \frac{K P_A^2 P_D}{(1 + K_A P_A + K_B P_B)^2}}$$